

Test 33: L-scaling ϵ_r (10 $\times$ 20 $\times$ 30 $\times$ 50 $\times$ 80 $\times$ 112)

Verdict: PASS | Generated: 2026-09-15 07:26:31 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

Statistical finite-size trend: monotone approach to 0.594-0.599; monograph anchors L $\geq$ 50, extended points L=80/112 (user spec 2026-08-29) check the GUE plateau; W_{eff} capped

METHOD: Statistical finite-size trend: monotone approach to 0.594-0.599; monograph anchors L $\geq$ 50, extended points L=80/112 (user spec 2026-08-29) check the GUE plateau; W_{eff} capped

Test 33 HARDCORE pass 2: $n \geq 5$ per L, weighted GUE-plateau ϵ_r over L $\geq$ 50, L capped at 112 (RAM honesty).

WHY W (DISORDER) IS CAPPED AT $W_{\text{max}} \geq 1.0 \geq$ disorder ϵ_r vortex motion ϵ_r statistics:

W is the diagonal on-site potential the vortices paint onto the lattice:

$$V_i = \sum_k q_k W / (r_{ik}^2 + 1), \quad \text{hopping } t = 1.$$

The ratio (disorder energy $q \bullet W$) / (kinetic energy t) decides what the

vortex cloud does and which spectral statistics come out:

$\epsilon_r q \bullet W \geq t$ (e.g. $W=4$, $q=0.3 \Rightarrow V \approx 1.2$ next to a core): the disorder landscape dominates the vortex motion ϵ_r vortices pin/steer the eigenstates, states localize on the potential relief, and ϵ_r drifts from GUE (0.5992) toward Poisson (0.3863) by an essentially random amount. A strongly disordered, spinning vortex system can therefore emit ALMOST ANY statistics ϵ_r the numbers depend on the effective rotation speed (W/t) and on the twisting/vortex arrangement (N_v , layout, charges) of that realization, not on the AB physics.

$\epsilon_r q \bullet W \geq t$ ($W \geq 1$): the Aharonov-Bohm phases dominate the hopping, states stay delocalized and GUE (Wigner-Dyson) is reachable.

That is why these tests measure at $W_{\text{eff}} = \min(\text{cfg.ab_}W, W_{\text{max}} = 1.0)$: they certify the PHASE physics, not the disorder physics.

Test 33 RAM gate: L=112 skipped (needs $\geq$ 10.6 GB peak, free $\geq$ 4.3 GB).

Extended L-scan to L=80 (user spec 2026-08-29); $W_{\text{eff}}=1.00$; verdict $q=0.30$; trend 0.434 $\geq$ 0.556 $\geq$ 0.593 $\geq$ 0.6 $\geq$ 0.6

Test 33 HARDCORE plateau ($q=0.30$): weighted $\epsilon_r=0.6004$ vs GUE ($|\epsilon_r|=0.0012$, in band), $\epsilon_r/dof=0.01$.

Test 33b (hardcore): plateau weighted $\epsilon_r=0.6004$, $\epsilon_r/dof=0.01 \geq$ PASS

- 'plots/plot_01.png' $\geq$ 1600 $\times$ 1000, ABPlotV23 engine (supersampled)

- 'plots/animation.gif' $\geq$ animated reveal of every plot of this test

Result

Plots

Independent verification guide

Reproduce with: `julia ab_cloud_v23.jl --test 33 --no-two-pass`

Every computation is logged in `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

FULL COMPUTATION LOG

[07:09:25.448] [+26556.476s] build_ab_cloud_hamiltonian: 10x10 (N=100), ?=0.5000, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[07:09:25.985] [+26557.014s] calc: ?r? = 0.629404 (n_r=59, n_eigs=61) [refs: GUE=0.5996 Poisson=0.3863]

[07:09:27.180] [+26558.208s] build_ab_cloud_hamiltonian: 10x10 (N=100), ?=0.5000, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[07:09:27.738] [+26558.766s] calc: ?r? = 0.509721 (n_r=59, n_eigs=61) [refs: GUE=0.5996 Poisson=0.3863]

[07:09:28.759] [+26559.788s] build_ab_cloud_hamiltonian: 10x10 (N=100), ?=0.5000, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[07:09:29.359] [+26560.388s] calc: ?r? = 0.574634 (n_r=59, n_eigs=61) [refs: GUE=0.5996 Poisson=0.3863]

[07:09:30.483] [+26561.512s] build_ab_cloud_hamiltonian: 10x10 (N=100), ?=0.5000, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[07:09:31.005] [+26562.034s] calc: ?r? = 0.557955 (n_r=59, n_eigs=61) [refs: GUE=0.5996 Poisson=0.3863]

[07:09:32.182] [+26563.211s] build_ab_cloud_hamiltonian: 10x10 (N=100), ?=0.5000, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[07:09:32.709] [+26563.737s] calc: ?r? = 0.486070 (n_r=59, n_eigs=61) [refs: GUE=0.5996 Poisson=0.3863]

[07:09:33.836] [+26564.864s] build_ab_cloud_hamiltonian: 20x20 (N=400), ?=0.5000, t=1.00, W=1.00, torus, 12 vortices, model=:monumental

[07:09:34.457] [+26565.485s] calc: ?r? = 0.593495 (n_r=239, n_eigs=241) [refs: GUE=0.5996 Poisson=0.3863]

[07:09:35.559] [+26566.587s] build_ab_cloud_hamiltonian: 20x20 (N=400), ?=0.5000, t=1.00, W=1.00, torus, 12 vortices, model=:monumental

[07:09:36.176] [+26567.204s] calc: ?r? = 0.596397 (n_r=239, n_eigs=241) [refs: GUE=0.5996 Poisson=0.3863]

[07:09:37.301] [+26568.330s] build_ab_cloud_hamiltonian: 20x20 (N=400), ?=0.5000, t=1.00, W=1.00, torus, 12 vortices, model=:monumental

[07:09:37.915] [+26568.943s] calc: ?r? = 0.582880 (n_r=239, n_eigs=241) [refs: GUE=0.5996 Poisson=0.3863]

[07:09:38.976] [+26570.004s] build_ab_cloud_hamiltonian: 20x20 (N=400), ?=0.5000, t=1.00, W=1.00, torus, 12 vortices, model=:monumental

[07:09:39.565] [+26570.593s] calc: ?r? = 0.588530 (n_r=239, n_eigs=241) [refs: GUE=0.5996 Poisson=0.3863]

[07:09:40.649] [+26571.677s] build_ab_cloud_hamiltonian: 20x20 (N=400), ?=0.5000, t=1.00, W=1.00, torus, 12 vortices, model=:monumental

[07:09:41.247] [+26572.275s] calc: ?r? = 0.615185 (n_r=239, n_eigs=241) [refs: GUE=0.5996 Poisson=0.3863]

[07:09:42.368] [+26573.396s] build_ab_cloud_hamiltonian: 30x30 (N=900), ?=0.5000, t=1.00, W=1.00, torus, 26 vortices, model=:monumental

[07:09:43.297] [+26574.326s] calc: ?r? = 0.625316 (n_r=539, n_eigs=541) [refs: GUE=0.5996 Poisson=0.3863]

[07:09:44.620] [+26575.649s] build_ab_cloud_hamiltonian: 30x30 (N=900), ?=0.5000, t=1.00, W=1.00, torus, 26 vortices, model=:monumental

[07:09:45.950] [+26576.979s] calc: ?r? = 0.610951 (n_r=539, n_eigs=541) [refs: GUE=0.5996 Poisson=0.3863]

[07:09:47.154] [+26578.182s] build_ab_cloud_hamiltonian: 30x30 (N=900), ?=0.5000, t=1.00, W=1.00, torus, 26 vortices, model=:monumental

[07:09:48.060] [+26579.089s] calc: ?r? = 0.600825 (n_r=539, n_eigs=541) [refs: GUE=0.5996 Poisson=0.3863]

[07:09:49.257] [+26580.285s] build_ab_cloud_hamiltonian: 30x30 (N=900), ?=0.5000, t=1.00, W=1.00, torus, 26 vortices, model=:monumental

[07:09:50.171] [+26581.199s] calc: ?r? = 0.592813 (n_r=539, n_eigs=541) [refs: GUE=0.5996 Poisson=0.3863]

[07:09:51.350] [+26582.378s] build_ab_cloud_hamiltonian: 30x30 (N=900), ?=0.5000, t=1.00, W=1.00, torus, 26 vortices, model=:monumental

[07:09:52.289] [+26583.317s] calc: ?r? = 0.605293 (n_r=539, n_eigs=541) [refs: GUE=0.5996 Poisson=0.3863]

[07:09:53.558] [+26584.586s] build_ab_cloud_hamiltonian: 50x50 (N=2500), $\gamma=0.5000$, $t=1.00$, $W=1.00$, torus, 70 vortices, model=:monumental

[07:09:59.555] [+26590.583s] calc: $\gamma=0.597483$ ($n_r=1499$, $n_{\text{eigs}}=1501$) [refs: GUE=0.5996 Poisson=0.3863]

[07:10:00.777] [+26591.806s] build_ab_cloud_hamiltonian: 50x50 (N=2500), $\gamma=0.5000$, $t=1.00$, $W=1.00$, torus, 70 vortices, model=:monumental

[07:10:07.510] [+26598.538s] calc: $\gamma=0.594415$ ($n_r=1499$, $n_{\text{eigs}}=1501$) [refs: GUE=0.5996 Poisson=0.3863]

[07:10:08.681] [+26599.709s] build_ab_cloud_hamiltonian: 50x50 (N=2500), $\gamma=0.5000$, $t=1.00$, $W=1.00$, torus, 70 vortices, model=:monumental

[07:10:15.270] [+26606.298s] calc: $\gamma=0.606851$ ($n_r=1499$, $n_{\text{eigs}}=1501$) [refs: GUE=0.5996 Poisson=0.3863]

[07:10:16.422] [+26607.450s] build_ab_cloud_hamiltonian: 50x50 (N=2500), $\gamma=0.5000$, $t=1.00$, $W=1.00$, torus, 70 vortices, model=:monumental

[07:10:22.981] [+26614.010s] calc: $\gamma=0.574458$ ($n_r=1499$, $n_{\text{eigs}}=1501$) [refs: GUE=0.5996 Poisson=0.3863]

[07:10:24.135] [+26615.163s] build_ab_cloud_hamiltonian: 50x50 (N=2500), $\gamma=0.5000$, $t=1.00$, $W=1.00$, torus, 70 vortices, model=:monumental

[07:10:30.667] [+26621.696s] calc: $\gamma=0.594551$ ($n_r=1499$, $n_{\text{eigs}}=1501$) [refs: GUE=0.5996 Poisson=0.3863]

[07:10:32.257] [+26623.285s] build_ab_cloud_hamiltonian: 80x80 (N=6400), $\gamma=0.5000$, $t=1.00$, $W=1.00$, torus, 178 vortices, model=:monumental

[07:12:09.993] [+26721.021s] calc: $\gamma=0.598524$ ($n_r=3839$, $n_{\text{eigs}}=3841$) [refs: GUE=0.5996 Poisson=0.3863]

[07:12:11.752] [+26722.781s] build_ab_cloud_hamiltonian: 80x80 (N=6400), $\gamma=0.5000$, $t=1.00$, $W=1.00$, torus, 178 vortices, model=:monumental

[07:13:46.304] [+26817.332s] calc: $\gamma=0.593096$ ($n_r=3839$, $n_{\text{eigs}}=3841$) [refs: GUE=0.5996 Poisson=0.3863]

[07:13:47.923] [+26818.951s] build_ab_cloud_hamiltonian: 80x80 (N=6400), $\gamma=0.5000$, $t=1.00$, $W=1.00$, torus, 178 vortices, model=:monumental

[07:15:21.288] [+26912.316s] calc: $\gamma=0.598961$ ($n_r=3839$, $n_{\text{eigs}}=3841$) [refs: GUE=0.5996 Poisson=0.3863]

[07:15:22.875] [+26913.903s] build_ab_cloud_hamiltonian: 80x80 (N=6400), $\gamma=0.5000$, $t=1.00$, $W=1.00$, torus, 178 vortices, model=:monumental

[07:16:55.922] [+27006.950s] calc: $\gamma=0.602712$ ($n_r=3839$, $n_{\text{eigs}}=3841$) [refs: GUE=0.5996 Poisson=0.3863]

[07:16:57.478] [+27008.506s] build_ab_cloud_hamiltonian: 80x80 (N=6400), $\gamma=0.5000$, $t=1.00$, $W=1.00$, torus, 178 vortices, model=:monumental

[07:18:29.703] [+27100.731s] calc: $\gamma=0.600199$ ($n_r=3839$, $n_{\text{eigs}}=3841$) [refs: GUE=0.5996 Poisson=0.3863]

[07:18:30.972] [+27102.000s] build_ab_cloud_hamiltonian: 10x10 (N=100), $\gamma=0.5000$, $t=1.00$, $W=1.00$, open, 4 vortices, model=:monumental

[07:18:31.457] [+27102.486s] calc: $\gamma=0.512863$ ($n_r=59$, $n_{\text{eigs}}=61$) [refs: GUE=0.5996 Poisson=0.3863]

[07:18:32.458] [+27103.487s] build_ab_cloud_hamiltonian: 10x10 (N=100), $\gamma=0.5000$, $t=1.00$, $W=1.00$, open, 4 vortices, model=:monumental

[07:18:32.939] [+27103.967s] calc: $\gamma=0.408227$ ($n_r=59$, $n_{\text{eigs}}=61$) [refs: GUE=0.5996 Poisson=0.3863]

[07:18:33.951] [+27104.979s] build_ab_cloud_hamiltonian: 10x10 (N=100), $\gamma=0.5000$, $t=1.00$, $W=1.00$, open, 4 vortices, model=:monumental

[07:18:34.430] [+27105.459s] calc: $\gamma=0.434605$ ($n_r=59$, $n_{\text{eigs}}=61$) [refs: GUE=0.5996 Poisson=0.3863]

[07:18:35.446] [+27106.474s] build_ab_cloud_hamiltonian: 10x10 (N=100), $\gamma=0.5000$, $t=1.00$, $W=1.00$, open, 4 vortices, model=:monumental

[07:18:35.926] [+27106.955s] calc: $\gamma=0.444497$ ($n_r=59$, $n_{\text{eigs}}=61$) [refs: GUE=0.5996 Poisson=0.3863]

[07:18:36.908] [+27107.936s] build_ab_cloud_hamiltonian: 10x10 (N=100), $\gamma=0.5000$, $t=1.00$, $W=1.00$, open, 4 vortices, model=:monumental

[07:18:37.391] [+27108.419s] calc: $\gamma=0.371493$ ($n_r=59$, $n_{\text{eigs}}=61$) [refs: GUE=0.5996 Poisson=0.3863]

[07:18:38.417] [+27109.445s] build_ab_cloud_hamiltonian: 20x20 (N=400), $\gamma=0.5000$, $t=1.00$, $W=1.00$, open, 12 vortices, model=:monumental

[07:18:38.972] [+27110.001s] calc: $\gamma=0.552303$ ($n_r=239$, $n_{\text{eigs}}=241$) [refs: GUE=0.5996 Poisson=0.3863]

[07:18:39.950] [+27110.978s] build_ab_cloud_hamiltonian: 20x20 (N=400), $\gamma=0.5000$, $t=1.00$, $W=1.00$, open, 12 vortices, model=:monumental

[07:18:40.491] [+27111.519s] calc: $\gamma=0.568758$ ($n_r=239$, $n_{\text{eigs}}=241$) [refs: GUE=0.5996 Poisson=0.3863]

[07:18:41.473] [+27112.501s] build_ab_cloud_hamiltonian: 20x20 (N=400), ?=0.5000, t=1.00, W=1.00, open, 12 vortices, model=:monumental

[07:18:42.014] [+27113.042s] calc: ?r? = 0.525701 (n_r=239, n_eigs=241) [refs: GUE=0.5996 Poisson=0.3863]

[07:18:42.989] [+27114.017s] build_ab_cloud_hamiltonian: 20x20 (N=400), ?=0.5000, t=1.00, W=1.00, open, 12 vortices, model=:monumental

[07:18:43.530] [+27114.558s] calc: ?r? = 0.549972 (n_r=239, n_eigs=241) [refs: GUE=0.5996 Poisson=0.3863]

[07:18:44.497] [+27115.525s] build_ab_cloud_hamiltonian: 20x20 (N=400), ?=0.5000, t=1.00, W=1.00, open, 12 vortices, model=:monumental

[07:18:45.034] [+27116.062s] calc: ?r? = 0.585633 (n_r=239, n_eigs=241) [refs: GUE=0.5996 Poisson=0.3863]

[07:18:46.009] [+27117.037s] build_ab_cloud_hamiltonian: 30x30 (N=900), ?=0.5000, t=1.00, W=1.00, open, 26 vortices, model=:monumental

[07:18:46.826] [+27117.854s] calc: ?r? = 0.578164 (n_r=539, n_eigs=541) [refs: GUE=0.5996 Poisson=0.3863]

[07:18:47.833] [+27118.861s] build_ab_cloud_hamiltonian: 30x30 (N=900), ?=0.5000, t=1.00, W=1.00, open, 26 vortices, model=:monumental

[07:18:48.633] [+27119.661s] calc: ?r? = 0.588849 (n_r=539, n_eigs=541) [refs: GUE=0.5996 Poisson=0.3863]

[07:18:49.616] [+27120.644s] build_ab_cloud_hamiltonian: 30x30 (N=900), ?=0.5000, t=1.00, W=1.00, open, 26 vortices, model=:monumental

[07:18:50.409] [+27121.438s] calc: ?r? = 0.583585 (n_r=539, n_eigs=541) [refs: GUE=0.5996 Poisson=0.3863]

[07:18:51.385] [+27122.414s] build_ab_cloud_hamiltonian: 30x30 (N=900), ?=0.5000, t=1.00, W=1.00, open, 26 vortices, model=:monumental

[07:18:52.173] [+27123.201s] calc: ?r? = 0.601887 (n_r=539, n_eigs=541) [refs: GUE=0.5996 Poisson=0.3863]

[07:18:53.154] [+27124.183s] build_ab_cloud_hamiltonian: 30x30 (N=900), ?=0.5000, t=1.00, W=1.00, open, 26 vortices, model=:monumental

[07:18:53.934] [+27124.962s] calc: ?r? = 0.614023 (n_r=539, n_eigs=541) [refs: GUE=0.5996 Poisson=0.3863]

[07:18:54.923] [+27125.951s] build_ab_cloud_hamiltonian: 50x50 (N=2500), ?=0.5000, t=1.00, W=1.00, open, 70 vortices, model=:monumental

[07:19:00.339] [+27131.367s] calc: ?r? = 0.606783 (n_r=1499, n_eigs=1501) [refs: GUE=0.5996 Poisson=0.3863]

[07:19:01.309] [+27132.338s] build_ab_cloud_hamiltonian: 50x50 (N=2500), ?=0.5000, t=1.00, W=1.00, open, 70 vortices, model=:monumental

[07:19:06.911] [+27137.940s] calc: ?r? = 0.602431 (n_r=1499, n_eigs=1501) [refs: GUE=0.5996 Poisson=0.3863]

[07:19:07.834] [+27138.862s] build_ab_cloud_hamiltonian: 50x50 (N=2500), ?=0.5000, t=1.00, W=1.00, open, 70 vortices, model=:monumental

[07:19:13.282] [+27144.310s] calc: ?r? = 0.597144 (n_r=1499, n_eigs=1501) [refs: GUE=0.5996 Poisson=0.3863]

[07:19:14.269] [+27145.298s] build_ab_cloud_hamiltonian: 50x50 (N=2500), ?=0.5000, t=1.00, W=1.00, open, 70 vortices, model=:monumental

[07:19:19.681] [+27150.710s] calc: ?r? = 0.608952 (n_r=1499, n_eigs=1501) [refs: GUE=0.5996 Poisson=0.3863]

[07:19:20.646] [+27151.674s] build_ab_cloud_hamiltonian: 50x50 (N=2500), ?=0.5000, t=1.00, W=1.00, open, 70 vortices, model=:monumental

[07:19:26.240] [+27157.269s] calc: ?r? = 0.586730 (n_r=1499, n_eigs=1501) [refs: GUE=0.5996 Poisson=0.3863]

[07:19:27.445] [+27158.474s] build_ab_cloud_hamiltonian: 80x80 (N=6400), ?=0.5000, t=1.00, W=1.00, open, 178 vortices, model=:monumental

[07:20:53.366] [+27244.395s] calc: ?r? = 0.597930 (n_r=3839, n_eigs=3841) [refs: GUE=0.5996 Poisson=0.3863]

[07:20:54.979] [+27246.007s] build_ab_cloud_hamiltonian: 80x80 (N=6400), ?=0.5000, t=1.00, W=1.00, open, 178 vortices, model=:monumental

[07:22:17.065] [+27328.093s] calc: ?r? = 0.610283 (n_r=3839, n_eigs=3841) [refs: GUE=0.5996 Poisson=0.3863]

[07:22:18.526] [+27329.554s] build_ab_cloud_hamiltonian: 80x80 (N=6400), ?=0.5000, t=1.00, W=1.00, open, 178 vortices, model=:monumental

[07:23:39.386] [+27410.415s] calc: ?r? = 0.595793 (n_r=3839, n_eigs=3841) [refs: GUE=0.5996 Poisson=0.3863]

[07:23:40.806] [+27411.834s] build_ab_cloud_hamiltonian: 80x80 (N=6400), ?=0.5000, t=1.00, W=1.00, open, 178 vortices, model=:monumental

[07:25:06.015] [+27497.044s] calc: ?r? = 0.599605 (n_r=3839, n_eigs=3841) [refs: GUE=0.5996 Poisson=0.3863]

[07:25:07.597] [+27498.625s] build_ab_cloud_hamiltonian: 80x80 (N=6400), ?=0.5000, t=1.00, W=1.00, open, 178 vortices, model=:monumental

[07:26:29.007] [+27580.036s] calc: ?r? = 0.598340 (n_r=3839, n_eigs=3841) [refs: GUE=0.5996 Poisson=0.3863]

CONSOLE CAPTURE

??

TEST 33 (pass 2, HARDCORE): L-scaling ? deep validation

Harder machinery: up to 5 realizations per L, weighted

GUE-plateau ?? over L ? 50, no L beyond 112 (RAM honesty: L=128 ? 4.3 GB).

??

? RUNTIME: 2 q-rows ×up to n=5 realizations ×L ? {10,20,30,50,80,112} ? expect ? 27?81 min (−3×machine factor; RAM-gated rungs are skipped inside with a printed reason)

??

TEST 33: L-Scaling for AB-Cloud Lattice

L=10?0.487 (Poisson), L=20?0.562, L=30?0.595, L=50?0.594 (monograph),

L=80, L=112 ? GUE plateau check (extended range, user spec 2026-08-29)

??

WHY W (DISORDER) IS CAPPED AT W_max ? 1.0 ? disorder ? vortex motion ? statistics:

W is the diagonal on-site potential the vortices paint onto the lattice:

$$V_i = \sum_k q_k W / (r_{ik} \cdot N_{\cdot} + 1), \quad \text{hopping } t = 1.$$

The ratio (disorder energy $q \cdot W$) / (kinetic energy t) decides what the

vortex cloud does and which spectral statistics come out:

? $q \cdot W$? t (e.g. $W=4$, $q=0.3$? $V \approx 1.2$ next to a core): the disorder

landscape dominates the vortex motion ? vortices pin/steer the

eigenstates, states localize on the potential relief, and ?r? drifts

from GUE (0.5992) toward Poisson (0.3863) by an essentially random

amount. A strongly disordered, spinning vortex system can therefore

emit ALMOST ANY statistics ? the numbers depend on the effective

rotation speed (W/t) and on the twisting/vortex arrangement (N_v ,

layout, charges) of that realization, not on the AB physics.

? $q \cdot W$? t ($W \geq 1$): the Aharonov-Bohm phases dominate the hopping,

states stay delocalized and GUE (Wigner-Dyson) is reachable.

That is why these tests measure at $W_{\text{eff}} = \min(\text{cfg.ab_}W, W_{\text{max}} = 1.0)$:

they certify the PHASE physics, not the disorder physics.

? RAM GATE: L=112 needs ? 10.6 GB peak (matrix + LAPACK workspace), free ? 4.3 GB ? SKIPPED (a swap-thrashing eig is indistinguishable from a hang)

q=1.00 BC=:torus L=10 Nv=4 ?=0.50 W=1.00 ?r? = 0.5516 – 0.0252 (5 real, monograph 0.487)

q=1.00 BC=:torus L=20 Nv=12 ?=0.50 W=1.00 ?r? = 0.5953 – 0.0055 (5 real, monograph 0.562)

q=1.00 BC=:torus L=30 Nv=26 ?=0.50 W=1.00 ?r? = 0.6070 – 0.0054 (5 real, monograph 0.595)

q=1.00 BC=:torus L=50 Nv=70 ?=0.50 W=1.00 ?r? = 0.5936 – 0.0053 (5 real, monograph 0.594)

q=1.00 BC=:torus L=80 Nv=178 ?=0.50 W=1.00 ?r? = 0.5987 – 0.0016 (5 real, monograph 0.595*)

q=0.30 BC=:open L=10 Nv=4 ?=0.50 W=1.00 ?r? = 0.4343 – 0.0233 (5 real, monograph 0.487)

q=0.30 BC=:open L=20 Nv=12 ?=0.50 W=1.00 ?r? = 0.5565 – 0.0100 (5 real, monograph 0.562)

q=0.30 BC=:open L=30 Nv=26 ?=0.50 W=1.00 ?r? = 0.5933 – 0.0065 (5 real, monograph 0.595)

q=0.30 BC=:open L=50 Nv=70 ?=0.50 W=1.00 ?r? = 0.6004 – 0.0040 (5 real, monograph 0.594)

q=0.30 BC=:open L=80 Nv=178 ?=0.50 W=1.00 ?r? = 0.6004 – 0.0025 (5 real, monograph 0.595*)

Verdict run: q=0.30 (fractional AB phase ? reference standard, stable)

Trend (L=10?20?30?50?80): 0.434 ? 0.556 ? 0.593 ? 0.6 ? 0.6

Monotone increase: YES

Final L=80 close to GUE 0.595 (|?|<0.05): YES

(comparison) q=1.0 user setting: 0.552 ? 0.595 ? 0.607 ? 0.594 ? 0.599 (final 0.5987)

Test 33: L-scaling to L=80 0.434,0.556,0.593,0.6,0.6 ? PASS

??? HARDCORE GUE-plateau check (verdict q=0.3, L ? 50) ???

plateau points: L=50: 0.6004–0.0040, L=80: 0.6004–0.0025

weighted ?r? = 0.6004 vs GUE 0.5992 (|?|=0.0012) ? IN ? band –0.022

?? = 0.01, dof = 2, ??/dof = 0.01 ? consistent ? (? 2 consistent; diagnostic)

Test 33b (hardcore): plateau weighted ?r?=0.6004, ??/dof=0.01 ? PASS